Adaptive Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

Aditya Dendukuri

University of California, Santa Barbara

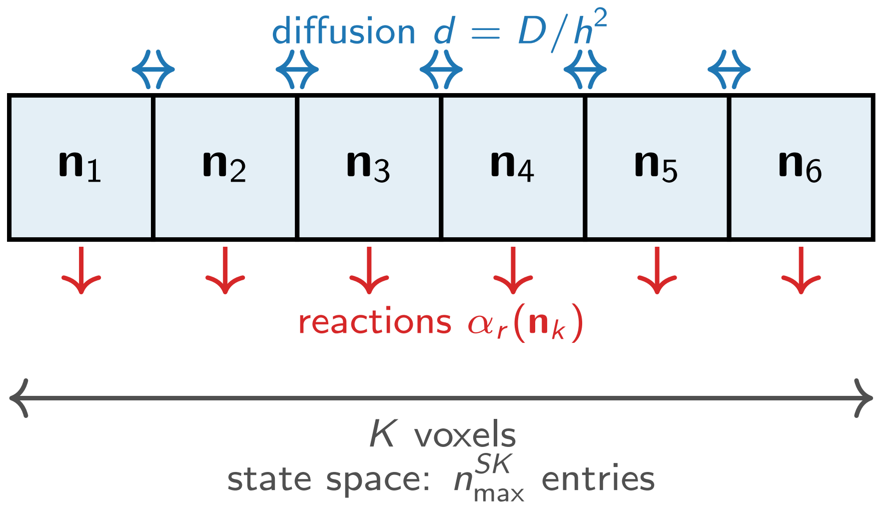
The RDME: reactions and diffusion across a spatial grid

Model. Divide a 1-D domain into K voxels. Each voxel k holds molecule counts $\mathbf{n}_k \in \mathbb{Z}^S_{\geq 0}$.

Two types of events:

- **Reactions** inside voxel k: rate $\alpha_r(\mathbf{n}_k)$, change $\mathbf{n}_k$.
- **Diffusion** between neighbors: rate $d = D/h^2$, moves one molecule $k \leftrightarrow k+1$.

The joint probability $p(\mathbf{n}_1, \ldots, \mathbf{n}_K, t)$ satisfies $\dot{p} = Ap$ **(RDME)**.



The computational bottleneck

The state space has $\sim n_{\max}^{SK}$ reachable states — **exponential in** K. Even $K{=}8$, $S{=}1$ is intractable with a naive FSP.

The probability ODE $\dot{p} = Ap$ has block-tridiagonal structure

$$\begin{array}{cc} & \begin{array}{cccc}\mathbf{n}_1 & \mathbf{n}_2 & \mathbf{n}_3 & \mathbf{n}_4\end{array} \\ \begin{array}{c}\mathbf{n}_1\\ \mathbf{n}_2\\ \mathbf{n}_3\\ \mathbf{n}_4\end{array} & \begin{bmatrix} A_1 & D_{12} & & \\ D_{12} & A_2 & D_{23} & \\ & D_{23} & A_3 & D_{34} \\ & & D_{34} & A_4 \end{bmatrix} \end{array} \cdot \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix} = \begin{bmatrix} \dot{p}_1 \\ \dot{p}_2 \\ \dot{p}_3 \\ \dot{p}_4 \end{bmatrix}$$

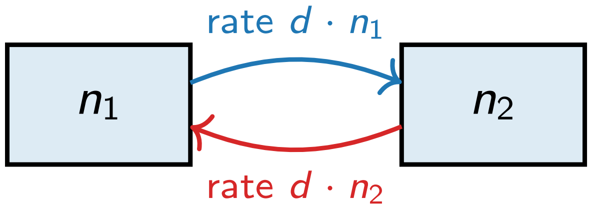
each block: $n_{\max}^{S} \times n_{\max}^{S}$ total: $n_{\max}^{SK} \times n_{\max}^{SK}$

A_k: reactions inside voxel k $D_{k,k+1}$: diffusion between neighboring voxels p_k: probability sub-vector for voxel k's state

Fast diffusion equilibrates two adjacent voxels

Fix total molecules $\bar{n} = n_1 + n_2$ across two adjacent voxels.

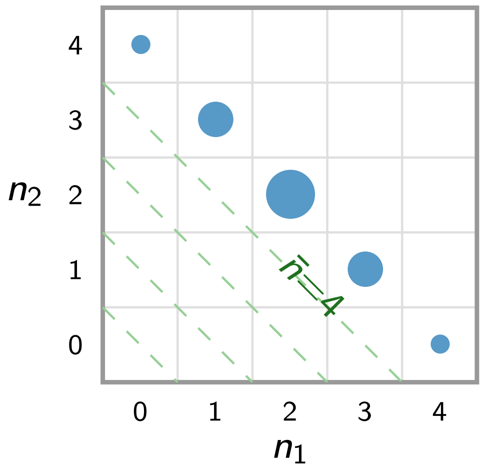
Diffusion moves one molecule at a time, at rate d per molecule:



Each molecule hops independently, so at equilibrium it is equally likely to be in either voxel. For $\bar{n}$ total:

$$\boxed{P(n_1 \mid \bar{n}) = \mathrm{Bin}\big(\bar{n}, \tfrac{1}{2}\big)} \quad \text{(exact, any } d > 0)$$

When $d \gg$ reactions: the two voxels are near this equilibrium and $\bar{n}$ **alone describes their joint state**.



Probability is spread along each constant-total diagonal as $\mathrm{Bin}(\bar{n}, \frac{1}{2})$ (dot size $\propto$ mass).

The two-voxel state is captured by $\bar{n}$ alone.

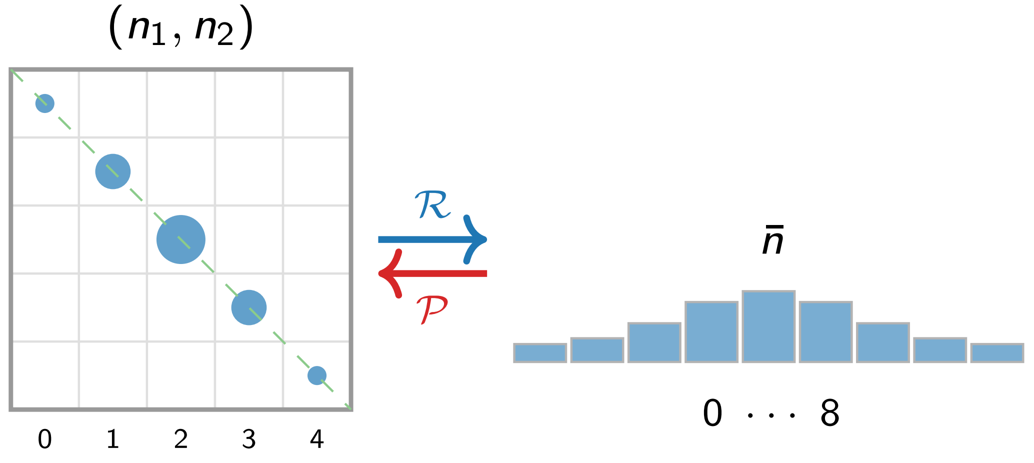
Grouping two adjacent voxels into their total

Compress ($\mathcal{R}$): sum over all splits with the same total:

$$p^c(\bar{n}) = \sum_{n_1+n_2=\bar{n}} p(n_1, n_2).$$

Reconstruct ($\mathcal{P}$): spread the total back via Binomial:

$$p(n_1, n_2) \approx p^c(\bar{n}) \cdot \mathrm{Bin}(\bar{n}, \tfrac{1}{2}).$$



Summing along constant-total diagonals (left) gives the 1-D total $\bar{n}$ (right). Reconstruction spreads mass back via Binomial.

State-space reduction

Per two voxels: $n_{\max}^{2S} \to n_{\max}^{S}$. All K voxels grouped: $n_{\max}^{SK} \to n_{\max}^{SK/2}$.

Square root of the original state space.

Valid when diffusion dominates between the two voxels. Fails when reactions prevent equilibration.

What coarsening does to the generator

Full generator A — 4 voxels:

$$
\begin{array}{c|cccc|}
 & \overbrace{n_1 \quad n_2}^{\text{group 1}} & & \overbrace{n_3 \quad n_4}^{\text{group 2}} & \\
\hline
n_1 & A_1 & D_{12} & & \\
n_2 & D_{12} & A_2 & D_{23} & \\
n_3 & & D_{23} & A_3 & D_{34} \\
n_4 & & & D_{34} & A_4 \\
\hline
\end{array}
$$

Size: $n_{\max}^4$

$$\xrightarrow{\mathcal{R}}$$

$$\xleftarrow{\mathcal{P}}$$

Coarsened generator $\bar{A}$ — 2 groups:

$$
\begin{array}{c|cc|}
 & \bar{n}_1 & \bar{n}_2 \\
\hline
\bar{n}_1 & \bar{A}_1 & \bar{D}_{12} \\
\bar{n}_2 & \bar{D}_{12} & \bar{A}_2 \\
\hline
\end{array}
$$

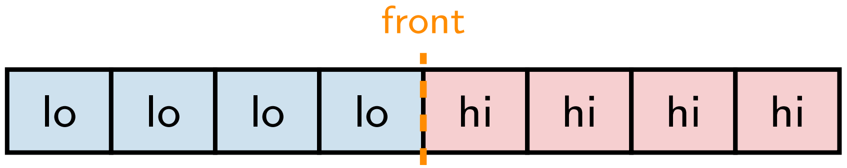
Size: $n_{\max}^2$ **($10^4\times$** smaller for $n_{\max}=100$)

- $A_k \to \bar{A}_j$: reactions averaged over the Binomial
- Within-group D disappears — implicit in the Binomial
- $\bar{D}_{12}$: only boundary diffusion between groups remains

Grouping fails at a bistable front

Schlögl model ($K{=}8$, $S{=}1$): each voxel has two stable states.

A **front** forms when neighboring voxels occupy opposite stable states. Slow diffusion ($d \ll$ reaction rates) sustains it indefinitely.



At the front: the joint distribution of the two interface voxels is *bimodal* — concentrated at $(n_{\mathrm{lo}}, n_{\mathrm{hi}})$ and $(n_{\mathrm{hi}}, n_{\mathrm{lo}})$.

The Binomial is **unimodal**: wrong shape.

Top: uniform initial condition — probability spreads along constant-total diagonals; Binomial reconstruction is accurate. ✓

Bottom: bistable front — probability stays at the corners; Binomial reconstruction gives the wrong shape. ✗

Detecting where grouping is valid: the imbalance criterion

Compare the mean counts in adjacent voxels $2j-1$ and $2j$:

$$\alpha_j = \frac{|\mu_{2j-1} - \mu_{2j}|}{\mu_{2j-1} + \mu_{2j}}.$$

- $\alpha_j \approx 0$: both voxels balanced $\Rightarrow$ diffusion dominates $\Rightarrow$ group them.
- $\alpha_j \approx 1$: voxels in opposite states $\Rightarrow$ front is here $\Rightarrow$ keep fine.

Grouping rule

Coarsen group j when $\alpha_j < \alpha_{\mathrm{lo}}$.
Keep fine when $\alpha_j > \alpha_{\mathrm{hi}}$.
(Two thresholds prevent flickering between states.)

(a) Birth-death: α_1 decays to zero as diffusion equilibrates the two voxels.
(b) Schlögl front: α_1 stays near 1 — the front is persistent.

Green band = safe region for grouping.

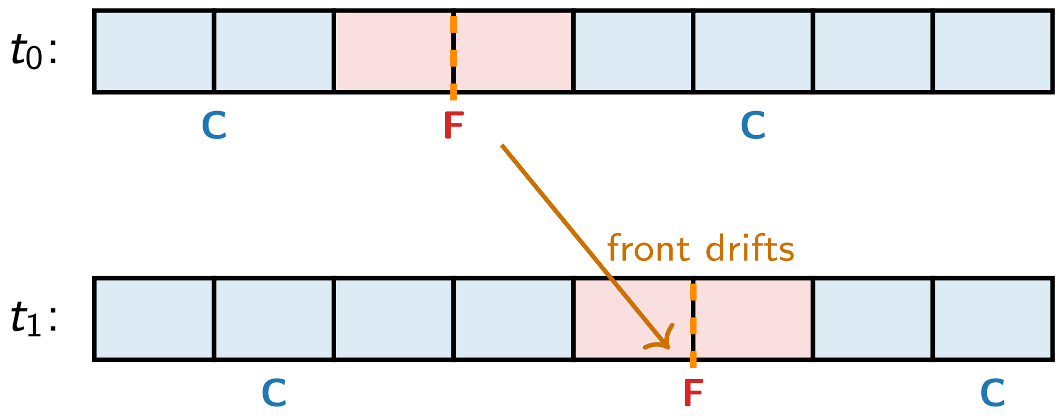
Idea. Keep a fine window at the front; coarsen everything else. Evolve the mixed state *directly* — no split-back step.

Mixed state p^M:

- **Coarse groups**: only the total $\bar{n}_j$ is tracked; split assumed Binomial.
- **Fine group at front**: both voxel counts (n_{2j-1}, n_{2j}) tracked exactly.

Evolution: $\dot{p}^M = A^M p^M$, where A^M couples coarse and fine blocks at group boundaries.

Reassignment: check α_j periodically; refine/coarsen only when the front crosses a group boundary — at most $K/2$ times total.



Fine window follows the front. Coarse regions stay compressed.

Adaptive mixed-resolution FSP: one time step

Algorithm 1 One step of the adaptive mixed-resolution FSP

Input: Mixed state p^M, partition (F, C), front position $\bar{\jmath}_{\mathrm{prev}}$, step Δt

1: **Solve:** $p^M \leftarrow e^{\Delta t\, A^M}\, p^M$ *(matrix exponentiation on the current mixed state)*

2: **Truncate:** drop states with $p^M < \tau$; expand by one-step reachability *(adaptive FSP)*

3: **Monitor:** $\bar{\jmath} \leftarrow \sum_j j\,(\mu_{2j-1} + \mu_{2j}) \,/\, \sum_j (\mu_{2j-1} + \mu_{2j})$ *(pair center of mass, $O(|\mathcal{S}^M|)$)*

4: **if** $|\bar{\jmath} - \bar{\jmath}_{\mathrm{prev}}| \geq \frac{1}{2}$ **then**

5: **Evaluate** $\alpha_j = |\mu_{2j-1} - \mu_{2j}| / (\mu_{2j-1} + \mu_{2j})$ for all groups j

6: **Update partition:** refine group j if $\alpha_j > \alpha_{\mathrm{hi}}$; coarsen if $\alpha_j < \alpha_{\mathrm{lo}}$; otherwise keep current assignment

7: **Transfer state:** newly fine groups via Binomial conditioning $\mathrm{Bin}(\bar{n}_j, \frac{1}{2})$; newly coarse groups via marginalization $\bar{n}_j = n_{2j-1} + n_{2j}$

8: $\bar{\jmath}_{\mathrm{prev}} \leftarrow \bar{\jmath}$

9: **end if**

Output: Updated p^M, partition (F, C), position $\bar{\jmath}_{\mathrm{prev}}$

Steps 1–2 run every step. Steps 4–8 (repartition) run only when the front crosses a group boundary — at most $K/2$ times total for a drifting front.

Results: adaptive criterion eliminates grouping error

Total variation distance from the reference (full FSP).
(a) Birth-death: grouping from $t=0$ gives large transient TV. Adaptive (coarsen only when $\alpha < \alpha_{\mathrm{lo}}$): TV $\to 0$ after equilibration.

(b) Schlögl front: naïve grouping gives TV ≈ 1 throughout (orientation of interface is lost). Adaptive: TV ≈ 0 throughout.

Results: adaptive mixed-resolution is accurate and fast

Means of each voxel: adaptive mixed-resolution vs. full FSP. Agreement to plotting accuracy.

Speedup vs. full FSP as K grows. Mixed-resolution scales well; full FSP becomes intractable.

Summary

1. Fast diffusion enables state-space compression.

When diffusion between two voxels dominates reactions, their joint distribution equilibrates to $\mathrm{Bin}(\bar{n}, \frac{1}{2})$. Replacing (n_1, n_2) by $\bar{n}$ cuts the per-group state space from $n_{\max}^{2S}$ to $n_{\max}^{S}$.

2. Bistable fronts violate the Binomial assumption.

Reactions keep neighboring voxels in opposite stable states, producing a bimodal conditional — not Binomial. The imbalance criterion α_j detects this automatically.

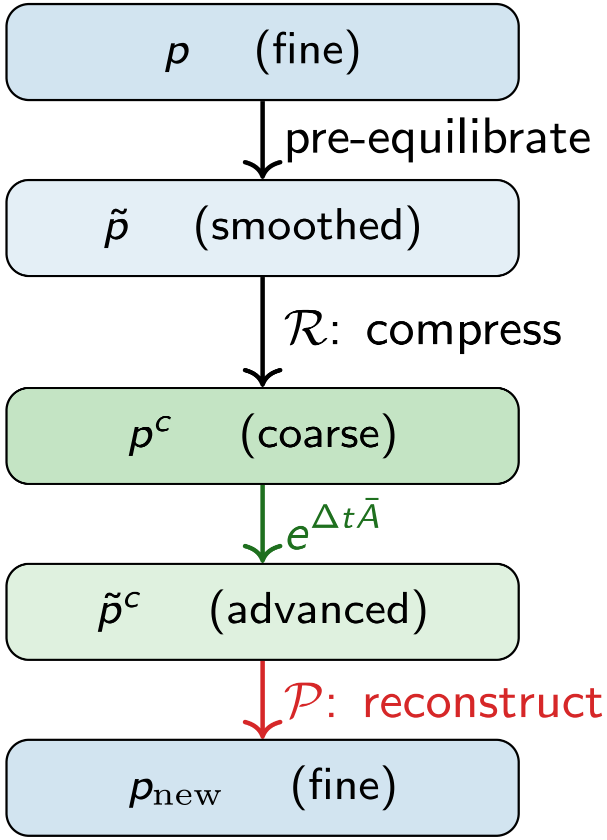
3. Adaptive mixed-resolution FSP.

Maintain a fine window at the front; coarsen the rest. Evolve $\dot{p}^M = A^M p^M$ directly — no reconstruction step. Reassign only when the front crosses a group boundary. **Accurate and tractable for K where full FSP is not.**

Open questions. Rigorous error bound replacing α_j; multi-species reactions; deeper hierarchies ($m > 2$).

Appendix: why not reconstruct after every step?

Naïve approach: pre-equilibrate → compress → coarse solve → reconstruct.



Reconstruction is wrong at the front

Step 4 forces a Binomial split on *every* group of adjacent voxels, including the front where the true distribution is bimodal.

Fix: evolve p^M directly — no reconstruction step. Coarse groups carry the Binomial assumption implicitly in $\bar{A}_j$; the fine group tracks the true within-group distribution exactly.

Appendix: mixed-resolution generator is self-consistent

Direct ODE solve on the mixed state $(\bar{n}_1, n_3, n_4)$ vs. Monte Carlo on the same generator. Means agree to $<1\%$; TV $\sim N^{-1/2}$ (sampling noise only, no systematic bias).

Appendix: resolving interface orientation ($K\!=\!8$)

IC: interface voxels in equal superposition of $(n_{\mathrm{lo}}, n_{\mathrm{hi}})$ and $(n_{\mathrm{hi}}, n_{\mathrm{lo}})$. **Coarse** $P(\bar{n}_2)$: single peak — orientation lost. **Fine** $P(n_3, n_4)$: two modes — mixed state retains the within-group structure.

References I

- Erban, Chapman, Maini (2007). Stochastic simulations of reaction-diffusion. *arXiv:0704.1908*.
- Munsky & Khammash (2006). Finite State Projection. *J. Chem. Phys.*, 124:044104.
- Briggs, Henson, McCormick (2000). *A Multigrid Tutorial*, 2nd ed. SIAM.
- Dendukuri, Chandrasekaran, Petzold (2025). Adaptive multigrid FSP for the RDME. *In preparation*.